

Physics

Answer 1: Inertia is the property of a body to resist changes in its state of rest or motion. Example: Passengers fall backward when a bus suddenly starts.

Answer 2: Acceleration is the rate of change of velocity with time. Its SI unit is meter per second squared (m/s^2).

Answer 3: The principle of conservation of energy states that energy can neither be created nor destroyed; it can only be transformed from one form to another.

Answer 4: Newton's Second Law states that the rate of change of momentum is directly proportional to the applied force. Formula: $F = ma$. Example: A football travels farther when kicked harder.

Answer 5: Reflection of light is the bouncing back of light rays from a surface. Laws: (1) Angle of incidence equals angle of reflection. (2) Incident ray, reflected ray, and normal lie in the same plane.